

1) Simplify the complex expressions

- | | | |
|--|---|--|
| a) i^5 | j) $i^6 + i^{16} + i^{26} + i^{36} + i^{46} + i^{56}$ | s) $\frac{1}{i} + \frac{1}{1+i} + \frac{1}{1-i}$ |
| b) i^{33} | k) $i \times i^2 \times i^3 \times \dots \times i^{99} \times i^{100}$ | t) $\frac{5-3i}{i} + (1+2i)^2$ |
| c) i^{148} | l) $(1+i)(-1+i) + (1+2i)^2$ | u) $\frac{1+2i}{3-4i} - \frac{2-i}{5i^3}$ |
| d) i^{999} | m) $\frac{2+4i}{1+i}(2-i)$ | v) $\frac{i^3 + i^2 + i}{2+i} + (1+i)^2$ |
| e) $i^3 + i^6 + i^9$ | n) $\frac{2-4i}{1+i} + (1+2i)^2 i^3$ | w) $\left(\frac{2-i}{1-3i}\right)^{-1} + \frac{i}{1-2i} \times i$ |
| f) $i^{10} + i^{20} + i^{30} + i^{40}$ | o) $\frac{i+1}{i} : \frac{i-1}{2i+3}$ | x) $(-1+5i) : \left(2 - \frac{3+i}{2+i}\right)$ |
| g) $i^{77} - i^{27} + i^{17} - i^{67}$ | p) $1 - \frac{i}{i+1} + (i-1)i$ | y) $5\left(\frac{3+4i}{2+i}\right)^{-1} \left(\frac{2}{i-2}\right)^2 + \frac{1}{i}$ |
| h) $i^2 - 2i^7 + 8i^3 - 6i + i^5$ | q) $\frac{1+i}{1-i} + \frac{1-i}{1+i}$ | z) $\frac{ 1+i }{1-i}$ |
| i) $i^5 + i^{10} + i^{15} + i^{20} + i^{25} + i^{30}$ | r) $\frac{2+i}{2-i} - \frac{2-i}{2+i}$ | Z) $\frac{ -1-i }{1+i\sqrt{3}}$ |

2) Find the absolute value of a complex number :

- | | | |
|---|--|---|
| a) i | i) $-2\sqrt{3} + 2i$ | r) $(1+i)^3$ |
| b) $2-3i$ | j) $5-i5\sqrt{3}$ | s) $(3+i\sqrt{2})(3-i\sqrt{2})$ |
| c) $1+2i$ | k) $1+i$ | t) $(1+i\sqrt{2})(2-3i)(3+i)$ |
| d) $-3-4i$ | l) $\frac{2+i}{3}$ | u) $0.3563 - 0.9343i$ |
| e) $-\sqrt{2} + i\sqrt{7}$ | m) $6+8i$ | v) $\sin \frac{\pi}{4} - i \cos \frac{\pi}{4}$ |
| f) $\frac{3}{5} - \frac{4}{5}i$ | n) $-\frac{\sqrt{3}}{2} + \frac{1}{2}i$ | w) $\frac{(1+i)^3}{(1-i)^5}$ |
| g) $-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$ | o) $1+i\sqrt{3}$ | x) $\frac{i^{10}-i}{2i+1}$ |
| h) $\frac{1}{2} + \frac{\sqrt{3}}{2}i$ | p) $(1-2i)(3+i)$ | y) $1+2i - \frac{2-5i}{3-i}$ |

3) Find the sum, difference and product of complex numbers x and y :

- | | | |
|--------------------------------------|-------------------------------------|--|
| a) $x = 2 + 5i, y = 3 - 2i$ | g) $x = -7 + 3i, y = -2 + i$ | m) $x = 8 - i, y = -6 - 9i$ |
| b) $x = 4 - i, y = 1 + 7i$ | h) $x = 4 + 2i, y = -2 + 5i$ | n) $x = i, y = -10 + 5i$ |
| c) $x = 1 + 2i, y = 1 + 4i$ | i) $x = 1 + i, y = 2 + i$ | o) $x = -2 + 7i, y = -3$ |
| d) $x = -2 + i, y = -3 - 2i$ | j) $x = 2 + i, y = 2 - i$ | p) $x = 12 + 15i, y = 16 - 22i$ |
| e) $x = -5 + 2i, y = -3 + 3i$ | k) $x = 1 - 2i, y = 3 + 7i$ | r) $x = 42 - 6i, y = -51 + 99i$ |
| f) $x = -2 - 2i, y = -1 + 3i$ | l) $x = 6 + 3i, y = -5 + 2i$ | s) $x = 1 + 2.3i, y = 4.5 - 0.2i$ |

4) Find the quotient of complex numbers :

- | | | |
|-----------------------------------|---|--|
| a) $\frac{3 + 5i}{2 - i}$ | g) $\frac{5 + 3i}{-2 - i}$ | m) $\frac{4 + 5i}{3 + 2i}$ |
| b) $\frac{1 - i}{3 + 2i}$ | h) $\frac{-4 - i}{-2 - 4i}$ | n) $\frac{3 + i}{1 - 2i}$ |
| c) $\frac{2 + 3i}{1 - 2i}$ | i) $\frac{7 - 2i}{-3 + 2i}$ | o) $(2 - 12i) : 4i$ |
| d) $\frac{3 + i}{3 - i}$ | j) $\frac{1 + i}{1 + 2i}$ | p) $\frac{1}{\sqrt{3} + i\sqrt{2}}$ |
| e) $\frac{1 - 2i}{2 + i}$ | k) $\frac{5 + i\sqrt{5}}{5 - i\sqrt{5}}$ | r) $\frac{2 + 3i}{-3 + 2i}$ |
| f) $\frac{-3 - i}{1 + i}$ | l) $\frac{1 + 2i}{3 + 4i}$ | s) $\frac{1 - 2i}{3 + 7i}$ |

5) Write a given complex number in the trigonometric form :

- | | | |
|------------------------|---|--|
| a) -5 | g) $-1-i\sqrt{3}$ | m) $-2+2i$ |
| b) $-7i$ | h) $5-i5\sqrt{3}$ | n) $\frac{1}{4}+\frac{1}{4}i$ |
| c) $7i$ | i) $-\sqrt{2}+i\sqrt{6}$ | o) $3-i\sqrt{3}$ |
| d) $-1+i$ | j) $-2-i\sqrt{12}$ | p) $\frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}i$ |
| e) $1-i$ | k) $\frac{3}{2}+\frac{\sqrt{3}}{2}i$ | r) $\frac{1+i}{1-i}$ |
| f) $\sqrt{3}+i$ | l) $-\sqrt{3}-i$ | s) $\frac{1}{1+i}+\frac{1}{-1+i}$ |

6) Write a given complex number in the algebraic form

- | | | |
|---|--|--|
| a) $\sqrt{2}(\cos 315^\circ + i \sin 315^\circ)$ | g) $10(\cos 50^\circ + i \sin 50^\circ)$ | m) $\frac{\sqrt{2}}{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$ |
| b) $5(\cos 210^\circ + i \sin 210^\circ)$ | h) $\sqrt{2}\left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4}\right)$ | n) $7(\cos 180^\circ + i \sin 180^\circ)$ |
| c) $2\left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2}\right)$ | i) $4\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)$ | o) $\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$ |
| d) $4\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$ | j) $7\sqrt{2}\left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4}\right)$ | p) $2e^{\frac{i\pi}{4}}$ |
| e) $2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$ | k) $10\sqrt{2}\left(\cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}\right)$ | r) $7e^{\frac{i\pi}{3}}$ |
| f) $\cos 135^\circ + i \sin 135^\circ$ | l) $2\left(\cos \frac{5\pi}{2} + i \sin \frac{5\pi}{2}\right)$ | s) $4e^{\frac{i\pi}{2}}$ |

7) Find the power of a complex number :

a) $(1+i)^4$

g) $(-1+i)^6$

m) $(1+i\sqrt{3})^{14}$

b) $(1-i)^8$

h) $(-2\sqrt{3}-2i)^{12}$

n) $\left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^9$

c) $(2+i)^3$

i) $(-\sqrt{3}-i)^8$

o) $\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i\right)^4$

d) $(\sqrt{2}+i)^4$

j) $(-3+3i)^4$

p) $\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right)^8$

e) $(1+i)^{10}$

k) $(1-i\sqrt{3})^{12}$

r) $\left(\frac{1-i}{1+i}\right)^{20}$

f) $(1-i\sqrt{3})^5$

l) $(-2+i\sqrt{12})^6$

s) $\left(\frac{\sqrt{3}}{2} - \frac{1}{2}i\right)^{100}$

8) Solve the complex equations :

a) $x^2 + 5 = 0$

i) $2x^2 - 11x + 14 = 0$

r) $x^3 + 1 = 0$

b) $2x^2 + 7 = 0$

j) $x^2 - 2ix - 5 = 0$

s) $x^4 - 1 = 0$

c) $3x^2 + 4 = 0$

k) $x^2 - (2+i)x - 1 + 7i = 0$

t) $27x^3 + 125 = 0$

d) $5x^2 - 4x + 1 = 0$

l) $x^2 - (3+2i)x + 5 + i = 0$

u) $x^4 + 1 = 0$

e) $x^2 - 4x + 6 = 0$

m) $\frac{1}{x-2} + \frac{1}{x-11} = \frac{1}{x-8}$

v) $\sqrt{x^2 + 8ix + 9} = -x + i$

f) $5x^2 - 6x + 2 = 0$

n) $\frac{52}{x-10} + 10 + x = \frac{52}{10-x}$

w) $\sqrt{x^2 + 8ix - 16} = 2x - 5i$

g) $x^2 - 2x + 5 = 0$

o) $x - \frac{1}{x} = \frac{2}{i} - \frac{i}{2}$

x) $x^2 + 9 + \sqrt{x^2 + 9} = 0$

h) $x^2 - 4x + 5 = 0$

p) $x + i = \frac{1}{x} + \frac{1}{i}$

y) $32x^5 - 1 = 0$